



MICROBIOLOGY

CHAPTER 20 : GRAM NEGATIVE



**Done by:
Sundos ali**

**Reference:
Slide and
Dr sereen notes**

Chapter 20: Gram Negative bacteria

🌸 Starting from this chapter, we're going to learn how to know what types of bacteria do we have in a certain sample. $\langle (\sim) \rangle$

→ In this chapter specifically we're going to follow the classification of Gram Negative bacteria .

- So, How do we start? (o' ▽` o)/

Suppose we took a sample of soil to know what types of bacteria live in it.

We'll follow these steps to identify them:

- Bring three types of media (blood agar, nutrient agar and selective media)
Grow the unknown bacteria on them and incubate for 24 hours
- Observe how many types of bacteria you have based on the colors and shapes of your colonies
For example, you got two colonies: White and Yellow
- Open two files

Name the first one	Name the second one
White colony	Yellow colony
- Do Gram test, biochemical tests and physiological analysis and each result should be recorded in these files
*Once we start studying one of the colonies, we forget about the other
(focus on one colony at a time!)
- Determine the shape of your bacteria (Cocci, Bacilli or spiral)

- Gram negative bacteria are divided into three groups based on their shape:

- A. Spiral gram -ve bacteria
- B. Helical/Vibrioid gram -ve bacteria
- C. Bacilli(rod)\Cocci gram -ve bacteria

A. Spiral shaped gram -ve bacteria

- Only ONE phylum have the spiral shaped bacteria: *Spirochaetes*
- They have two major characteristics:
 1. Their shape: slender, very long with flexible helical shape (may reach 100 μm)
 2. Their motility: creeping (crawling) motility due to a structure called an axial filament which is a complex of axial fibrils that are embedded in the membrane
- they consist of:
 - PF: Periplasmic flagella (AF: Axial Filament)
 - PC: protoplasmic cylinder
 - OS: outer sheath
 - IP: insertion pore
- They are Chemoheterotrophs

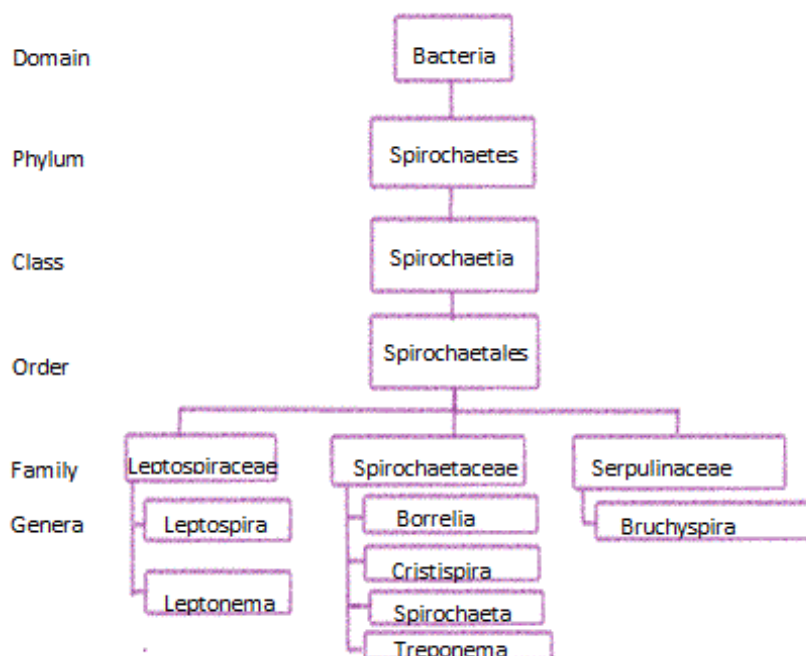
Energy source:

Inorganic
chemicals

Carbon
source:

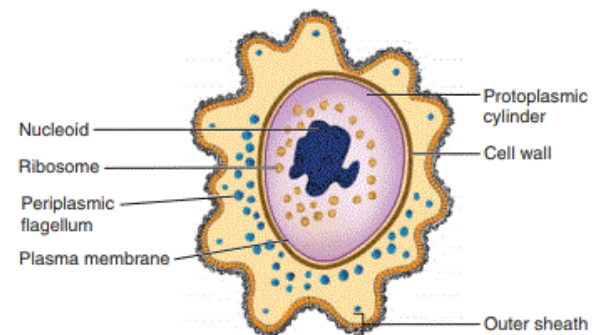
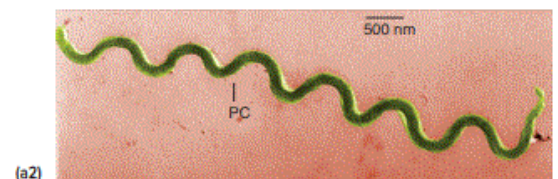
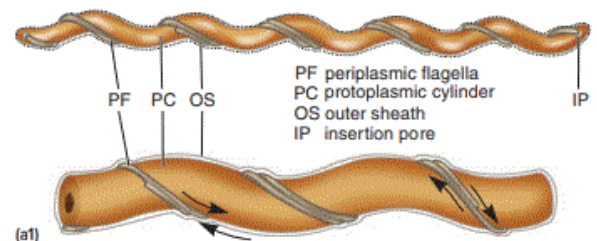
Organic
molecule

- Ecologically diverse
- Phylum Spirochaetes contains one class; one order, Spirochaetales ; three families, Spirochaetaceae, Serpulinaeae, and Leptospiraceae ; 13 genera



As a rule of Thumb!

ALL spiral shape bacteria are gram -ve

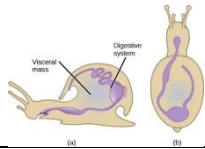


(b)

Figure 21.19 Spirochete Morphology. (a1) A surface view of spirochete structure as interpreted from electron micrographs. (a2) A longitudinal view of *Treponema pallidum*. (b) A cross section of a typical spirochete showing morphological details.

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Table 6.2 Sources of Carbon, Energy, and Electrons	
Carbon Sources	
Autotrophs	CO ₂ sole or principal biosynthetic carbon source (section 11.3)
Heterotrophs	Reduced, preformed, organic molecules from other organisms (chapters 10 and 11)
Energy Sources	
Phototrophs	Light (section 10.12)
Chemotrophs	Oxidation of organic or inorganic compounds (chapter 10)
Electron Sources	
Lithotrophs	Reduced inorganic molecules (section 10.11)
Organotrophs	Organic molecules (chapters 9 and 10)

Genus	Morphology	Oxygen requirement	Carbon and Energy source	Habitats
Spirochaeta	Periplasmic flagella	Facultatively anaerobic* or anaerobic	Carbohydrates	Aquatic and free living
Cristispira	Periplasmic flagella	Facultatively anaerobic	N.A†	Mollusk Digestive system 
Treponema	Periplasmic flagella	Anaerobic or microaerophilic**	Carbohydrates Or Amino acids	Mouth, intestinal tract, and genital areas of mammals; some are pathogenic (Syphilis, Yaws→Sexual diseases)
Borrelia	Periplasmic flagella	Anaerobic or microaerophilic	Carbohydrates	Mammals and arthropods; Highly pathogenic (Relapsing Fever ‡ Lyme disease) Hard to diagnose 
Bruchyspira	Periplasmic flagella	Anaerobic	N.A	Mammalian intestinal tract 
Leptonema	Periplasmic flagella	Aerobic	Fatty Acids	Mammals 
Leptospira	Periplasmic flagella	Aerobic	Fatty Acids and alcohols	Free living or pathogens for mammals located in the kidney (leptospirosis)
Serpulina	Periplasmic flagella	Anaerobic	Carbohydrates and amino acids	Mammalian intestinal tract

* Facultatively anaerobic: Live with or without oxygen, but it prefers the presence of oxygen

** **Microaerophilic: Requires oxygen in small quantities**

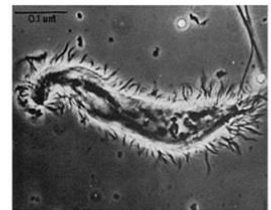
† N.A: information not available

‡ Relapsing fever: at a moment the body's temperature reaches 40 C and after a short while it reaches 37°C so it's hard to diagnose it ; the only way is by blood film

- Symbiotic Associations between Spirochetes and Other Organisms: between broad range of organisms found in a variety of locations, for example:

- hindguts of termites أحشاء النمل الأبيض
- digestive tracts of mollusks(الحلزونات) and mammals
- oral cavities of animals

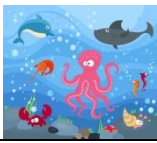
Spirochete-Protozoan Associations



(b) From S. C. Holt, Microbiological Reviews 42(1): 122, 1978 American Society for Microbiology

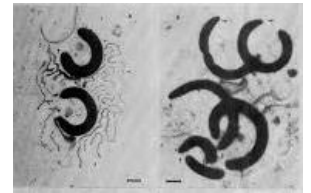
A. Helical\ Vibrioid grame -ve bacteria

- Quite diverse ecologically

Genus	Morphology	Oxygen Requirement	Habitats	Notes\Explanation
Spirillum	Large bipolar bundles of flagella	Microaerophilic	Stagnant (راكدة) fresh water	
Azospirillum	Single polar flagellum in liquid medium	Aerobic or Microaerophilic	Soil and associated with tree roots; can fix nitrogen 	- <u>Azo</u> : Means something related to nitrogen ALWAYS -They fix atmospheric nitrogen in the soil
Aquaspirillum	Usually helical; with bipolar tufts of flagella	Aerobic (a few may be Microaerophilic)	Fresh water	
Oceanspirillum	Helical; usually with bipolar tufts of flagella	Aerobic	Marine 	
Campylobacter	Vibrioid or Helical with a single flagellum at one or both poles	Microaerophilic	<u>Intestinal tract</u> , reproductive organs and oral cavity of humans and other animals (campylobacter enteritis)	-Highly pathogenic -can cause Septicimia: pathogen or their toxin in the blood -cause diarrhea for kids
Helicobacter	Helical, curved, or straight ;multiple sheathed flagella	Microaerophilic	Gastric mucosa of primates and ferrets  	-Highly pathogenic - the only bacteria that can survive in the stomach lining since it has an enzyme called Urease that breaks down urea inside the stomach juice transforming it into ammonia. ammonia reacts with HCl forming micronutral PH at the <u>surface</u> of the bacteria -causes ulcers and sometimes stomach cancer -80-90% of us have this bacteria but only 10-50% show symptoms -causes bad mouth smell, can be treated by taking antibiotics
Bdellovibrio	Vibrioid with a single sheathed polar flagellum	Aerobic	Soil, sewage, fresh water and marine; predaceous (تفترس) on other gram negative bacteria	-predatory bacteria: loves E.coli; sticks to it, forms pores in its cell wall and enters to live inside E.coli between the cell wall and the plasma membrane, reproduces there, kills E.coli and then attacks other E.coli

*All bacteria mentioned in the table are motile, gram -ve Helical\ Vibrioid bacteria

We only have one type that is NON-motile: Microcycclus → looks like an inverted C letter



➤ Slides (13,14,15)

- Genus *Campylobacter*

- contains both pathogenic and nonpathogenic species

○ *Campylobacter fetus*

- reproductive disease and abortions (اجهاضات) in cattle and sheep
- septicemia and enteritis (التهاب الأمعاء) in humans
 - septicemia - pathogens or their toxins in blood
 - enteritis - inflammation of intestinal tract

○ *Campylobacter jejuni*

- Abortions in sheep
- enteritis diarrhea in humans

It's so important to differentiate between these two (٢٠٢٠)

- Genus *Helicobacter*

- at least 14 species isolated from stomachs and upper intestines of humans, dogs, cats, and other mammals

- e.g., *Helicobacter pylori*

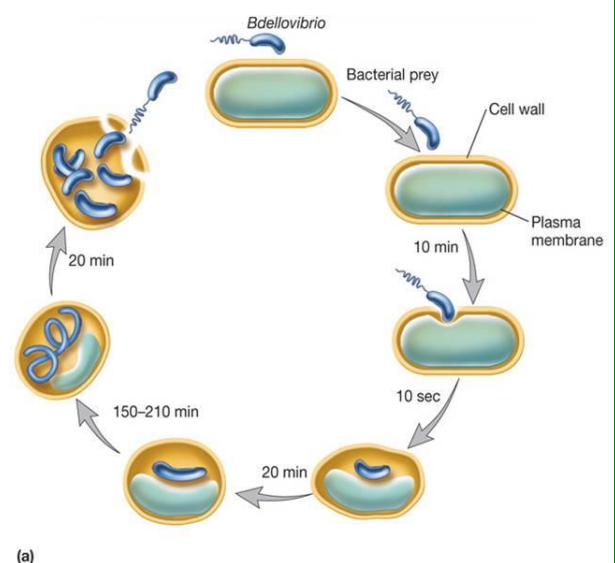
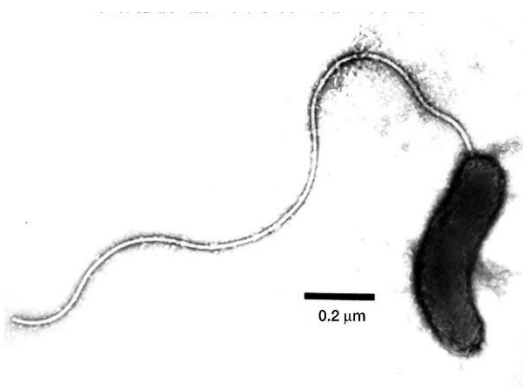
- causes gastritis and peptic ulcer disease
- produces large quantities of urease
 - urea hydrolysis appears to be associated with virulence

- Order Bdellovibrionales

- contains one family, *Bdellovibrionaceae*; four genera

○ best studied is *Bdellovibrio*

- **predatory bacteria**

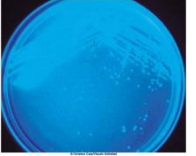


B. Bacilli\Cocci gram -ve bacteria

- All are strict aerobic : it requires oxygen highly

→ They form oxygen singlates which is toxic for any living cell so they must have catalase enzyme and based on that all the following genera have catalase enzyme

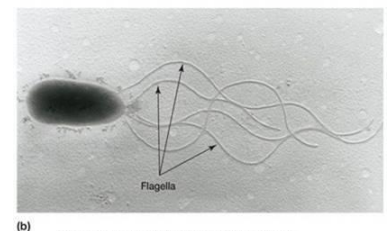
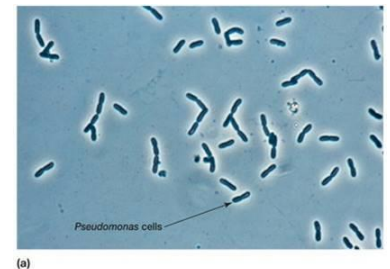
***Any bacterial name that ends with -bacter or -monas it means that it has a bacillus shape**

Family	Genera	Notes\Explination
Pseudomonadaceae	Burkholderia	Causes many diseases in intestines 
	Pseudomonas	Degrade more than 100 organic compound in the soil so they're very important → this process is called mineralization -Pseudomonas aeruginosa: have a blue-green pigmentation that causes problems in the middle ear (Otitis Media) 
	Xanthomonas	Secret a substance used in chocolate making 
	Zoogloea	-Very essential for moist water remediation -they have a large capsule
Azotobacteraceae	Azotobacter	Both do nitrogen fixation in the soil under aerobic conditions (they are like fertilizers) They do it freely in the soil meaning that they are NOT associated with plants
	Azomonas	
Rhizobiaceae	Rhizobium	-Rhizo: means roots -found on the roots of legumes(بقوليات) and they form nodules there -they live with plants symbiotically (both benefit) - Plants benefit from nitrogen that has been fixed by the bacteria - Bacteria takes nutrients from the plant 
	Agrobacterium	-important for plant biotechnology -they cause tumors in plants by invading the plant and transforming cells into proliferating tumor cells -contain a large Ti plasmid that is used by scientists to introduce certain characteristics to a plant by inserting new genes to it 
Methylococcaceae	Methylococcus	Methane oxidizing bacteria: they use methane and methanol as their sole carbon energy source
	Methylomonas	
Acetobacteraceae	Acetobacter	Oxidize ethanol and other alcohols and convert it to acetic acid
	Gluconobacter	

Legiononellaceae	Legionella	-Pathogenic: cause inflammation in trachea -They grow in moist environments	
Nesseriaceae	Neisseria	Highly pathogenic: like <i>Neisseria gonorrhoeae</i> that causes sexually transmitted diseases	
	Moraxella		
	Acinetobacter		

slides(19-32):

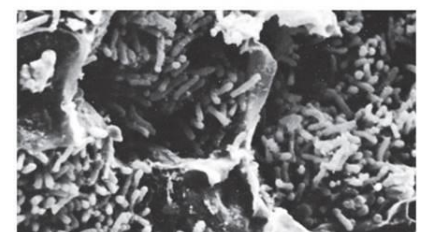
- Slide 19: Order *Pseudomonadales*
- contains family Pseudomonadaceae; 15 genera
 - *Pseudomonas* is the most important genus in the order Pseudomonadales
 - gram-negative straight or slightly curved rods
 - 0.5 to 1.0 mm by 1.5 to 5.0 mm in length
 - motile by one or several polar flagella
 - lack *prosthecae* or sheaths
- slide 20: *Pseudomonas*
- chemoheterotrophs with respiratory metabolism
 - usually use oxygen as electron acceptor
 - sometimes use nitrate as electron acceptor
 - have functional TCA cycle
 - most hexoses are degraded by Entner-Doudoroff pathway
- slide 21: Practical importance of *pseudomonads*:
 - metabolically versatile
 - degrade wide variety of organic molecules
 - mineralization: microbial breakdown of organic materials to inorganic substrates
 - important experimental subjects
 - some are major animal and plant pathogens
 - some cause spoilage of refrigerated food
 - can grow at 4°C
- slide24: Genus *Rhizobium*
- gram-negative, motile rods
 - often contain poly-β-hydroxybutyrate granules
 - become pleomorphic under adverse conditions
- grow symbiotically as nitrogen-fixing bacteroids within root nodule cells of legumes



ASM Microbiology.org. Photo micrograph by William Ghiorse; ASM Microbiology.org. Photo courtesy of Caroline Harwood, University of Iowa



(a)



(b)

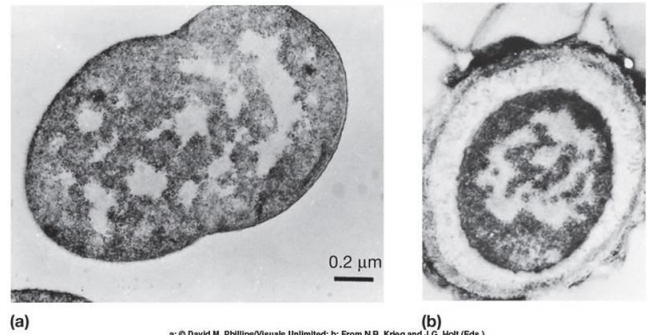
ASM Microbiology.org. Photo micrograph by William Ghiorse; ASM Microbiology.org. Photo courtesy of Caroline Harwood, University of Iowa

الدكتورة مرقت عن هاد السلايد

- slide 26: *Genus Agrobacterium*
- **do not** stimulate nodule formation or fix nitrogen
- invade crown, roots, and stems of many plants
 - transform infected plant cells into autonomously proliferating **tumors**
- e.g., *Agrobacterium tumefaciens*:
 - causes **crown gall disease** by means of tumor-inducing (Ti) plasmid

- slide 28: *Genus Azotobacter*

- chemoorganotrophs
- widespread in soil and water



- slide 30: *Order Methylococcales*

- contains family Methylococcaceae; seven genera
- morphologically diverse
 - e.g., genus *Methylococcus* - spherical, nonmotile
 - e.g., genus *Methylomonas* - straight, curved, or branched rods with single polar flagella
 - almost all form resting stage (cystlike structure)
- methylotrophs:
 - use reduced one-carbon compounds as sole carbon and energy source
- slide 31: Methane oxidation
 - occurs in complex arrays of intracellular membranes
 - oxidized to methanol and then to formaldehyde
 - electrons donated to electron transport chain for ATP synthesis
 - formaldehyde can be assimilated into cell material

مرقت عنهم slide30&slide31

- slide 32: *Genus Neisseria*

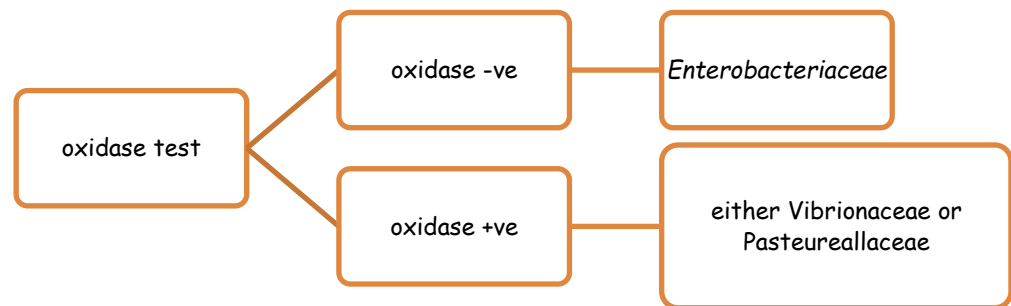
- nonmotile, gram-negative cocci
 - most often occur in pairs with adjacent sides flattened
 - may have capsules and fimbriae
- aerobic chemoorganotrophs
 - oxidase positive and usually catalase positive
- inhabitants of mucous membranes of mammals
 - some human pathogens: كثير مهمين
 - *Neisseria gonorrhoeae* - **gonorrhea**
 - *Neisseria meningitidis* - **meningitis**

- *Order Vibrionales*
- contains one family, *Vibrionaceae*; eight genera
- most are aquatic
 - o most free-living
 - o some important pathogens
 - o some symbiotic in luminous organs of fish
 - o Very fast motility (Shooting star)
- closely related to two other orders
 - o *Enterobacteriales* and *Pasteurellales*

→ o All of these (*Vibrionaceae*, *Enterobacteriaceae* and *Pasteurellaceae*) are all bacilli, facultative anaerobic gram -ve bacteria, So how can we differentiate between them?

- We follow those steps (ノ●▽●)/*:° ✧

1. Oxidase test



2. To differentiate between *Vibrionaceae* and *Pasteurellaceae* we do a Motility test

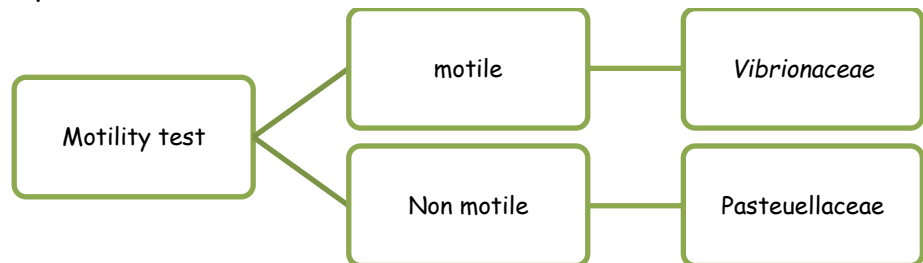
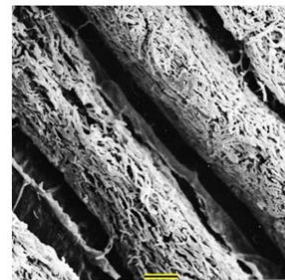


Table 22.6 Characteristics of Families of Facultatively Anaerobic Gram-Negative Rods

Characteristics	<i>Enterobacteriaceae</i>	<i>Vibrionaceae</i>	<i>Pasteurellaceae</i>
Cell dimensions	0.3–1.0 × 1.0–6.0 μm	0.3–1.3 × 1.0–3.5 μm	0.2–0.3 × 0.3–2.0 μm
Morphology	Straight rods; peritrichous flagella or nonmotile	Straight or curved rods; polar flagella	Cocci to rod-shaped cells, sometimes pleomorphic; nonmotile
Physiology	Oxidase negative	Oxidase positive; all can use D-glucose as sole or principal carbon source	Oxidase positive; heme and/or NAD often required for growth; organic nitrogen source required
G + C content	38–60%	38–63%	38–47%
Symbiotic relationships	Some parasitic on mammals and birds; some species plant pathogens	Most not pathogens (with a few exceptions)	Parasites of mammals and birds
Representative genera	<i>Escherichia</i> , <i>Shigella</i> , <i>Salmonella</i> , <i>Citrobacter</i> , <i>Klebsiella</i> , <i>Enterobacter</i> , <i>Erwinia</i> , <i>Serratia</i> , <i>Proteus</i> , <i>Yersinia</i>	<i>Vibrio</i> , <i>Photobacterium</i>	<i>Pasteurella</i> , <i>Haemophilus</i>

From J. G. Holt and N. R. Krieg (eds.), *Bergey's Manual of Systematic Bacteriology*, Vol. 1, Copyright © 1984 Williams and Wilkins Company, Baltimore, MD. Reprinted by permission.

- **Vibrio cholera**
 - causes cholera
 - genome has been sequenced
 - has two circular chromosomes (which is an exception)
 - copies of some genes present on both chromosomes
- **Vibrio fischeri**
 - capable of bioluminescence
 - emission of a blue green light catalyzed by luciferase {lives symbiotically with fish (on the eyes of squid)}
 - also observed in at least two species of *Photobacterium*
 - can use 3', 5'-cyclic AMP as source of C, N, and P
 - free-living



luminescent bacteria in luminous organs of their fish hosts

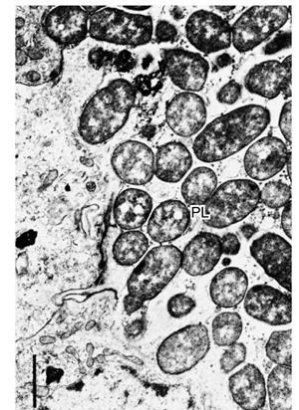
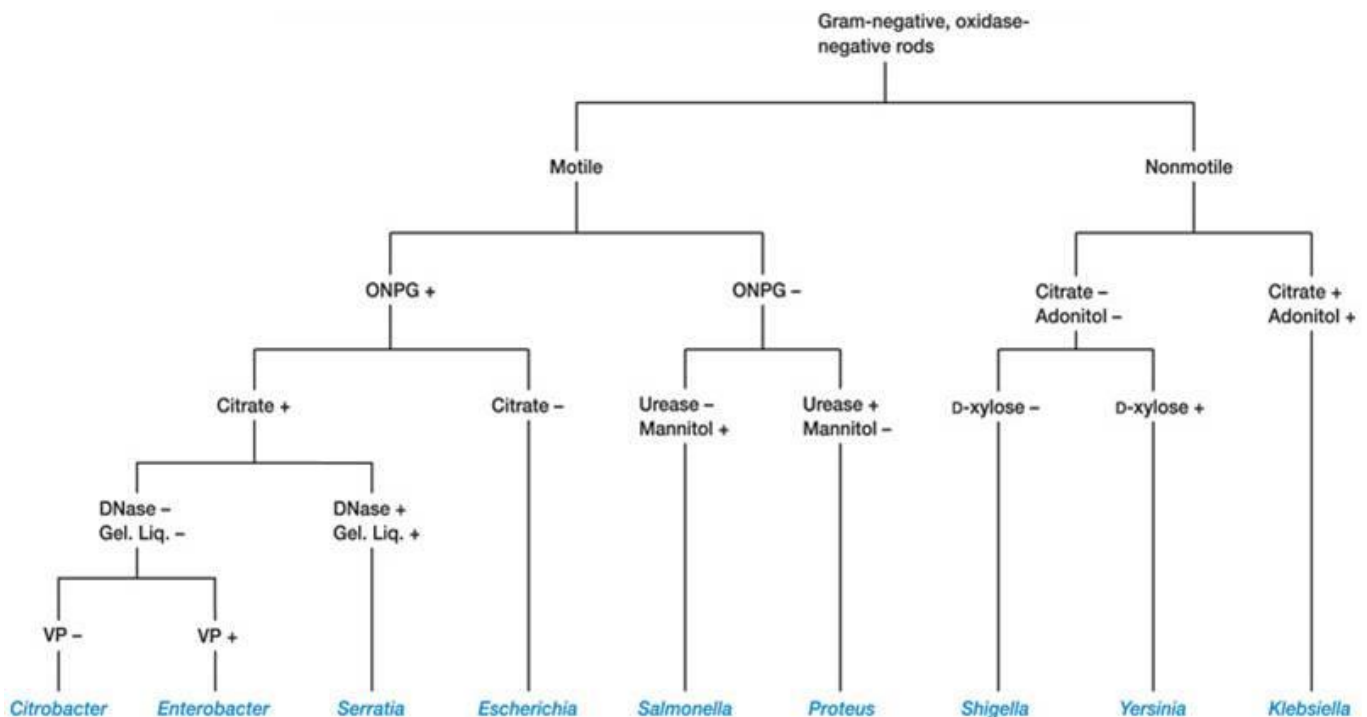
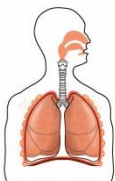
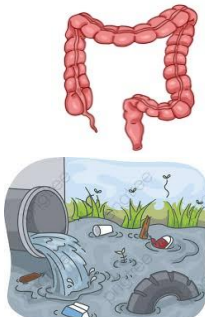


Figure 22.28 (b) and (c)

- Order Enterobacteriales
- contains one family, *Enterobacteriaceae*; over 44 genera
 - o referred to as enterobacteria or enteric bacteria (enterobacteria)
- facultative anaerobes
- chemoorganotrophs that degrade sugars by glycolytic pathway
 - o can cleave pyruvate to yield formic acid (formic acid fermentation)
- *the diagram represents the 9 genera of Enterobacteriaceae
- **ONPG: an enzyme called β -galactosidase that degrades lactose
- ***Gel.liq: gelatinase liquefaction
- ‡VP: test for sugar(glucose) fermentation



Genera	explanation
Shigella	<i>Shigella dysenteriae</i> causes diarrhea
Klebsiella	- Causes respiratory tract infection (pneumonia) - have a large capsule 
Yersinia	Causes plague (الطاعون)
Proteus	Causes urea tract infection
Salmonella	Salmonella Typhimurium: causes: 1. typhoid fever 2. food poisoning
Escherichia	- found in large intestine as normal flora - sewage water have billions of it (can be tested using Macconkey agar) 
Serratia	Makes jelly liquid by degrading gelatin 
Enterobacter	Very important Used to get rid of pollution
Citrobacter	Metabolizes citrate
Eriwina	- Major pathogen of crop plants - Causes wilting (ذبول) and other diseases - Can be found on fruits and vegetables and white cheese

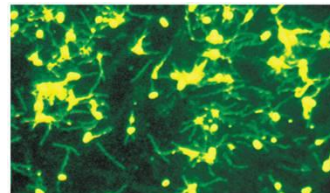
Characteristics	<i>Yersinia</i>	<i>Klebsiella</i>	<i>Enterobacter</i>	<i>Erwinia</i>	<i>Serratia</i>
Methyl red	+	(+) ^a	(-) ^b	+	d ^c
Voges-Proskauer	- (37°C)	(+)	+	(+)	+
Indole production	d	d	-	(-)	(-)
Citrate use	(-)	(+)	+	(+)	+
H ₂ S production	-	-	-	(+)	-
Urease	d	(+)	(-)	-	-
β-galactosidase	+	(+)	+	+	+

Table 22.7 Some Characteristics of Selected Genera in the *Enterobacteriaceae*

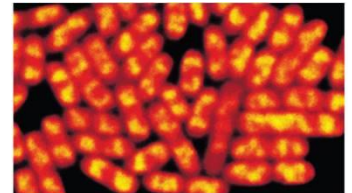
Characteristics	<i>Escherichia</i>	<i>Shigella</i>	<i>Salmonella</i>	<i>Citrobacter</i>	<i>Proteus</i>
Methyl red	+	+	+	+	+
Voges-Proskauer	-	-	-	-	d
Indole production	(+)	d	-	d	d
Citrate use	-	-	(+)	+	d
H ₂ S production	-	-	(+)	d	(+)
Urease	-	-	-	(+)	+
β-galactosidase	(+)	d	d	+	-

*You have to memorize them, simply memorize which are positive OR which are negative and you'll memorize the others by default $\neg(\neg\neg)\wedge(\neg\neg)/$

- *Escherichia coli*
 - probably best studied bacterium
 - inhabitant of intestinal tracts of many animals
 - used as indicator organisms for testing water for fecal contamination
 - some strains are pathogenic
 - gastroenteritis
 - urinary tract infections
- Important pathogenic enteric bacteria
 - *Salmonella* - typhoid fever and gastroenteritis
 - *Shigella* - bacillary dysentery
 - *Klebsiella* - pneumonia
 - *Yersinia* - plague
 - *Erwinia* - blights, wilts, etc., of crop plants



(a) *Salmonella enteritidis*



(b) *S. typhi*

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- Order *Pasteurellales*

- contains one family, Pasteurellaceae; six genera

- Important pathogens:

- *Pasteurella multocida* : fowl cholera
- *Pasteurella haemolytica* : pneumonia in cattle, sheep and goats
- *Haemophilus influenzae* : variety of diseases, including meningitis in children

→ this bacteria was discovered during the influenza pandemic and mistakenly was thought to be the cause of the influenza and as a result, it was named like this.

- Class *Bacteroides*

- gram-negative rods of various shapes
 - do not form endospores
 - motile or nonmotile
- anaerobic chemoheterotrophs
 - fermentative
- often found in oral cavity and intestinal tract of humans and other animals and the rumen of ruminants* as normal flora
 - often benefit host by degrading complex carbohydrates, providing extra nutrition to host
 - can constitute up to 30% of bacteria from human feces
 - some cause disease when moved from the intestines to the blood stream and also they cause pimples when cosmetics are not cleaned probably.

**Ruminants: are mammals that are able to acquire nutrients from plant-based food because they have a specialized stomach that have microbial actions*

- Orders *Desulfovibrionales*, *Desulfobacterales*, and *Desulfuromonadales*

- strict anaerobes
- sulfur- or sulfate-reducing bacteria
 - use sulfur and sulfate as electron acceptors during anaerobic respiration
 - electron transport chain used to generate ATP
- widespread in muds and sediments of aquatic environments, including sewage treatment systems
 - important in sulfur cycling

- mostly when you find black or dark regions near the sea or mud, you'll find these sulfur reducing bacteria (some are gram +ve and some are -ve)

→ Often sulfate reduction is apparent from the smell of Hydrogen Sulfide H_2S and the blackening of water and sediment by the precipitation of Iron Sulfide Fe_2S

- Notice how the name gives an illustration for the shape
Desulfovibrionales: vibrio
Desulfobacterales and *Desulfuromonadales*: bacilli
- *Desulfovibrionales* have a negative impact on industry because they do corrosions (تآكل) to iron, pipe lines, heating systems and other things and they ruin them.

- Related to Clostridia

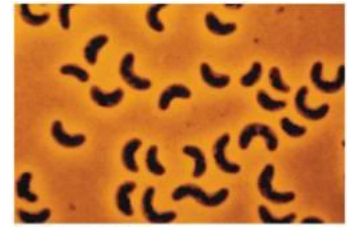
- Heliobacteria
 - Photosynthetic bacteria (have photosystem similar to green sulfur bacteria) it does photosynthesis under anaerobic conditions
- Veillonella
 - organisms gram-negative but placed in order Clostridiales
 - all have complex nutritional requirements
 - Anaerobic gram -ve cocci shaped bacteria
 - Found as normal flora in the gastrointestinal tract, urinary or genital tract and in the mouth but it can cause diseases according to its location

- Gram -ve very very small bacteria (☆~☆)

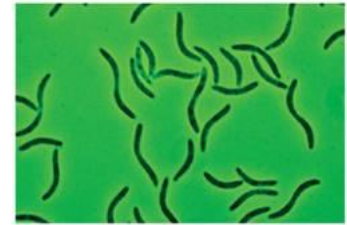
They are very small (as the size of the largest virus → Pox virus) and Pathogenic

→ 3 groups:

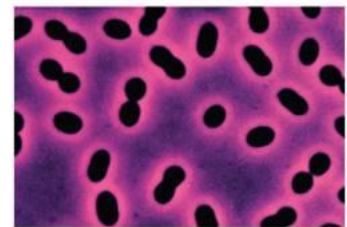
1. *Rickettsia* and *Coxiella*
2. *Chlamydia*
3. *Mycoplasma*



(a) *Desulfovibrio sapovorans*



(b) *Desulfovibrio gigas*



(c) *Desulfobacter postgatei*
a-c: © F. Widdel/Visuals Unlimited

1. *Rickettsia* and *Coxiella*

Genus *Rickettsia*

class Alphaproteobacteria; order Rickettsiales; family

- considered together because of important similarities

Genus *Coxiella*

class Gammaproteobacteria; order Legionellales; family Coxiellaceae

- rod-shaped, coccoid, or pleomorphic
 - typical gram-negative cell walls
 - no flagella
 - very small
 - *Rickettsia* - 0.3 to 0.5 by 0.8 to 2.0 μm
 - *Coxiella* - 0.2 to 0.4 by 0.4 to 1.0 μm
 - parasitic or mutualistic(متكامل)*
 - parasitic species grow in important cells inside the human: vertebrate erythrocytes, macrophages, and vascular endothelial cells
 - also live in blood-sucking arthropods (ticks, lice, etc..) which serve as vectors or primary hosts
- RBC's (erythrocytes) have an important role in regulating body temperature so many diseases caused by these bacteria have at the end of its name "fever"

**Mutualism describes the ecological interaction between two or more species where each species has a net benefit*

- Parasitic life styles $\leftarrow \sim(\Psi \nabla - \nabla) \in$

- Rickettsia*

The phagocytic cell forms pseudopodia, then it forms a vesicle around the bacteria and enters the cell. after that, *Rickettsia* escapes from the phagosome to the cytoplasm and reproduces there.

Rickettsia

enters host by phagocytosis



escapes phagosome



reproduces in cytoplasm



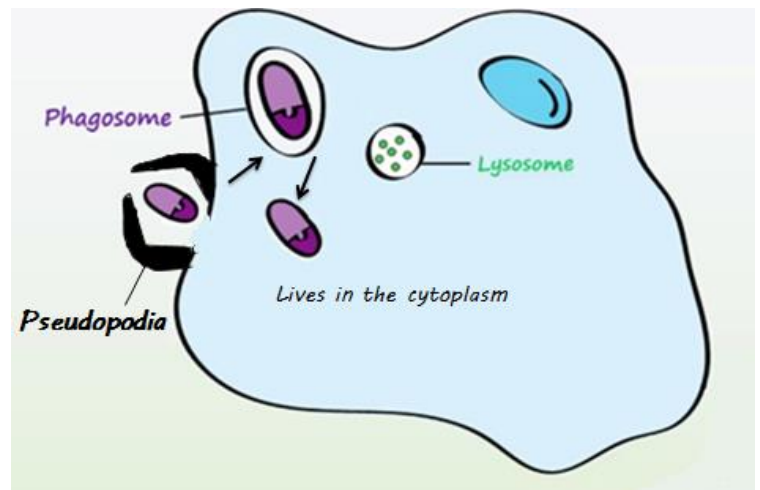
host cell bursts

- Metabolism:

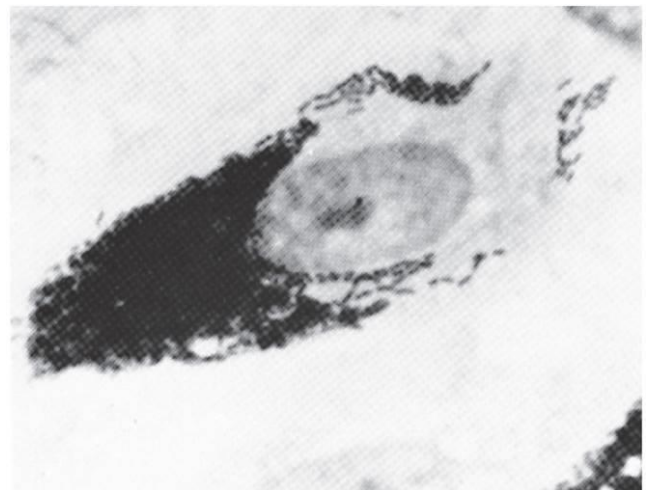
- lack glycolytic pathway
 - do not use glucose as energy source
 - oxidize glutamate and TCA cycle (Krebs cycle) intermediates (e.g., succinate)
 - take up and use ATP and other materials from host cell



it has its own metabolism but mostly it takes its energy from the host cell



Human
fibroblast
filled with
Rickettsia
proWazekii



(a)

From N.R. Krieg and J.G. Holt (Eds.), *Bergey's Manual of Systematic Bacteriology*, Vol. 1, 1984. Williams and Wilkins Co., Baltimore

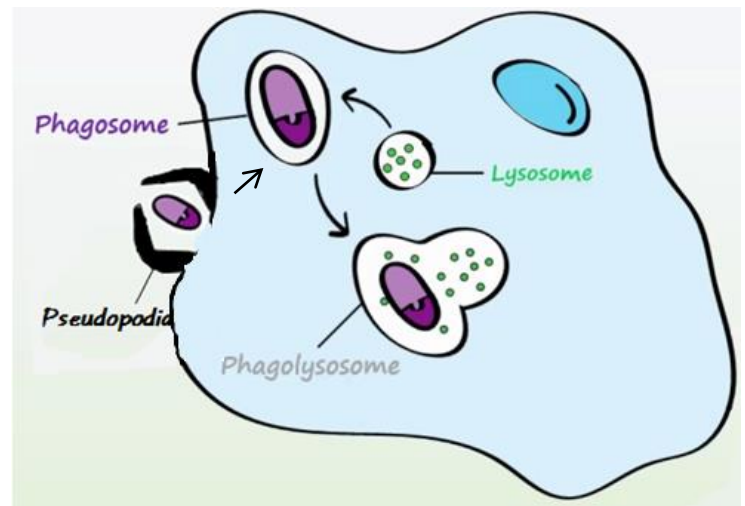
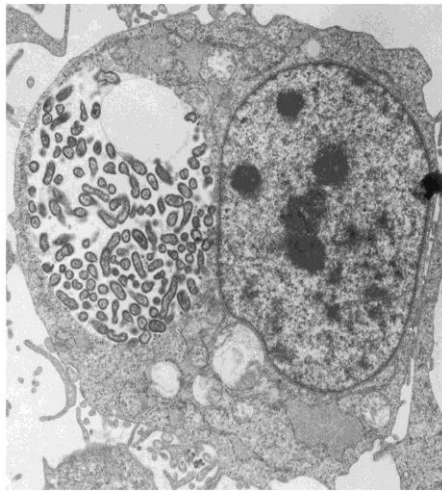
- *Coxiella*

Remains in the phagosome and then the lysosome units with phagosome forming a phagolysosome to break down the bacteria but *coxiella* won't be affected by those enzymes that are supposed to degrade the bacteria and reproduce there

Coxiella

enters host by phagocytosis
↓
remains in phagosome
↓
reproduces in phagolysosome
↓
host cell bursts

Coxiella
Burnetti
growing
within
fibroblast
vacuole



- Important pathogens

- Rickettsia prowazekii and Rickettsia typhi - typhus fever
- Rickettsia rickettsii - Rocky Mountain Spotted Fever (affects those who climb high mountains)
- Coxiella burnetii - Q fever: transmitted from animals to humans when drinking unpasteurized milk
- many are important pathogens in dogs, horses, sheep, and cattle

2. Phylum Chlamydiae

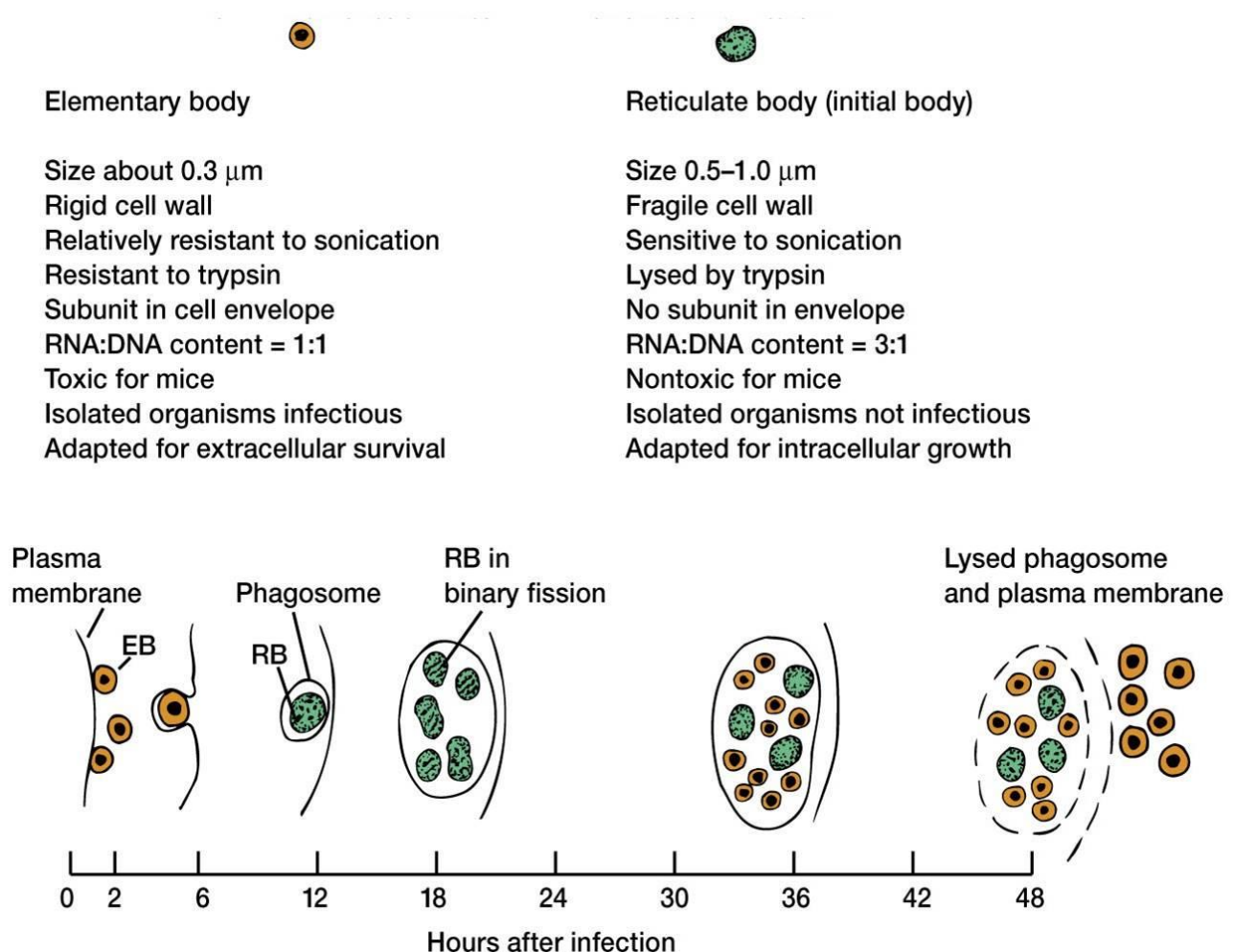
- one class, one order, four families and six genera
- nonmotile, coccoid, gram-negative bacteria
 - o cell walls lack muramic acid and peptidoglycan
 - o have very small genomes
- obligate intracellular parasites

although known for ability to cause disease, many grow within hosts such as protists, vertebrate and invertebrate cells without adverse effects.

(they enter our bodies through an intermediate host then they get engulfed through the phagocytic cell and live there)
- genus *Chlamydia* is best studied

➤ Have a unique developmental cycle

- involving formation of elementary body (EB) for attachment and reticulate body (RB) or initial body for reproduction
- found mostly in mammals and birds
- some recently isolated from spiders, clams, and freshwater invertebrates



- Chlamydial metabolism

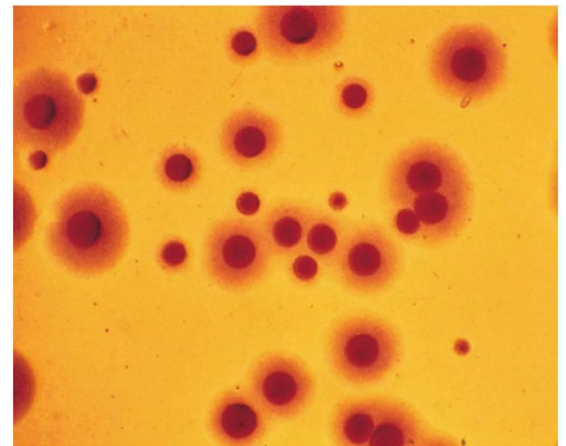
- appear to be energy parasites, obtaining ATP from host
 - do have genes for substrate-level phosphorylation, electron transport, and oxidative phosphorylation
- reticulate bodies have biosynthetic capabilities when supplied precursors from host; can synthesize some amino acids
- elementary bodies seem to be dormant forms

- Important pathogens

- *C. trachomatis*
 - infects humans and mice
 - causes trachoma, nongonococcal urethritis, and other diseases in humans
- *C. psittaci*
 - infects humans and many other animals
 - causes psittacosis in humans
- *C. pneumoniae*
 - common cause of human pneumonia

3. Class Mollicutes (The Mycoplasmas)

- contains five orders and six families
- lack cell walls and because of that its pleomorphic
 - cannot synthesize peptidoglycan precursors
 - Penicillin resistant: penicillin targets peptidoglycan and since mycoplasma does not have it, it won't be affected by it
 - it can be lysed using osmotic shock or detergents
- sterols may stabilize plasma membrane: its plasma membrane requires cholesterol to have a more rigid structure
 - So cholesterol must be added when we grow it in the lab and most species will form colonies with fried egg appearance
- most nonmotile; some have gliding motility
- smallest bacteria capable of self-reproduction
- genomes :
 - less than 1000 genes
 - one of the smallest found in procaryotes



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- Metabolism of Mycoplasmas
 - chemoorganotrophs:
 - some produce ATP by glycolysis and lactic acid fermentation
 - some catabolize amino acids and urea
 - some have functional pentose phosphate pathway
 - none have complete TCA cycle
 - deficient in a number of biosynthetic pathways
- Important pathogens
 - *Mycoplasma mycoides* - bovine pleuropneumonia in cattle
 - *Mycoplasma gallisepticum* - chronic respiratory disease in chickens
 - *Mycoplasma hyopneumoniae* - pneumonia in swine
 - *Mycoplasma pneumoniae* - primary atypical pneumonia in humans
 - *Ureaplasma urealyticum* - premature birth, neonatal meningitis and pneumonia
 - spiroplasmas - pathogenic in insects, ticks, and a variety of plants

The End

(~▽~)~ ~(~▽~)~